Experimental Characterization of Silicon-on-Insulator Interconnects: Cross-Validating Waveguide Dispersion via Linear Interferometry

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***Abstract*—This report details the theoretical modeling, simulation, and experimental characterization of silicon-on-insulator (SOI) photonic integrated circuits. The primary objective is the high-precision extraction of the waveguide group index (n_g) utilizing a series of unbalanced Mach-Zehnder Interferometers (MZI). Experimental data acquired from automated (Iris) and manual (Scylla) probing stations are cross-referenced with Finite Difference Eigenmode (FDE) simulations for 500×220 nm strip waveguides. Results yield an extracted $n_g \approx 4.21$, demonstrating excellent agreement with nominal simulations and confirming fabrication fidelity within predicted process corners.**

I. Introduction

Silicon-on-insulator (SOI) photonic integrated circuits enable high-index-contrast light confinement, supporting compact and highly dense interferometric architectures. However, the high modal confinement in SOI waveguides renders their phase response significantly sensitive to nanometer-scale geometric variations [**?**]. This sensitivity results in waveguide dispersion, necessitating accurate characterization of the effective index (n_{eff}) and group index (n_g).

Relying on a single test structure or measurement setup can introduce systematic measurement bias. Consequently, this research utilizes a parameterized array of linear MZIs alongside multi-station testing. The agreement between simulated and extracted n_g values provides independent verification of the dispersion model and isolates geometric variations from coupling-induced spectral shifts.

II. Waveguide Theory and Modal Analysis

A. FDE Solver and Light Confinement

Our standard waveguide is a silicon strip that is 500 nm wide and 220 nm tall, sitting on a $2\mu m$ layer of silicon dioxide (BOX). We utilized a Finite Difference Eigenmode (FDE) solver to compute the modal profile. As shown in Fig. 1, the fundamental TE mode is tightly confined inside the silicon core at $\lambda = 1550$ nm.

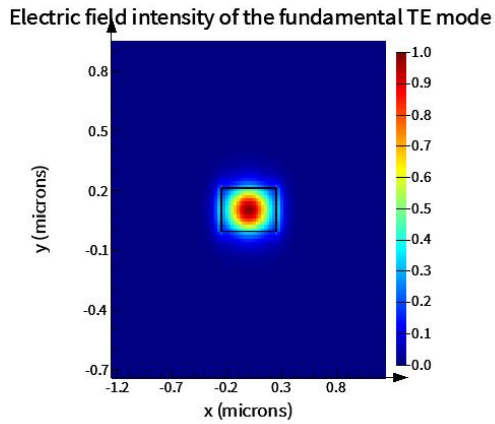

Fig. 1: Normalized electric field intensity ($|E|^2$) of the fundamental TE mode at $\lambda = 1550$ nm.

B. Dispersion and Group Index (n_g) Extraction

The effective index (n_{eff}) dictates the phase velocity of light, but due to chromatic dispersion, an optical envelope travels at the group index (n_g):

$$n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda} \tag{1}$$

Numerical analysis yields a nominal $n_g \approx 4.211$ at 1550 nm (Fig. 2). To account for manufacturing tolerances, a corner analysis was simulated (wafer thickness: 215.3–223.1 nm; waveguide width: 470–510 nm), establishing an acceptable n_g boundary of 4.16 to 4.24.

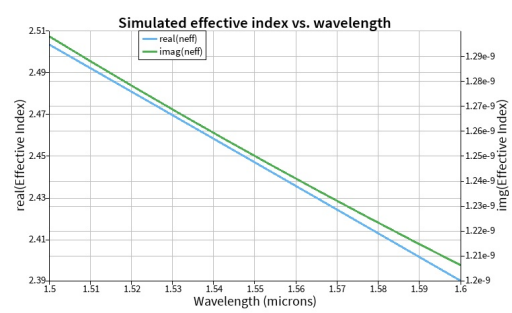

Fig. 2: Wavelength-dependent effective index (n_{eff}) simulation results.

C. Compact Model for Circuit Simulation

We fit the n_{eff} data to a second-order polynomial centered at $\lambda_0 = 1550$ nm:

$$n_{eff}(\lambda) \approx n_0 + n_1(\lambda - \lambda_0) + n_2(\lambda - \lambda_0)^2 \quad (2)$$

Coefficients: $n_0 = 2.446820$, $n_1 = -1.133390\mu m^{-1}$, and $n_2 = -0.043937\mu m^{-2}$.

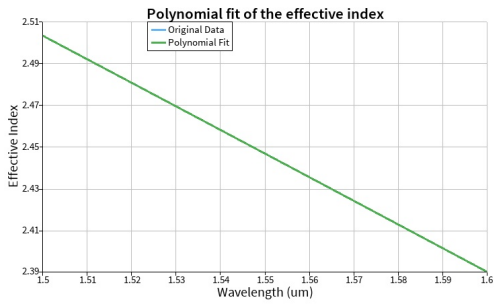

Fig. 3: Verification of the polynomial compact model fit.

III. Design of Interferometric Structures

A. Mach-Zehnder Interferometer (MZI) Theory

An MZI creates a phase shift ($\Delta\phi$) via a path length difference ΔL:

$$\Delta\phi = \frac{2\pi n_{eff}\Delta L}{\lambda}, \quad T(\lambda) = \frac{1}{2}\left[1 + \cos(\Delta\phi)\right] \quad (3)$$

The Free Spectral Range (FSR) is directly mapped to the group index:

$$FSR_{MZI} = \frac{\lambda^2}{n_g \Delta L} \quad (4)$$

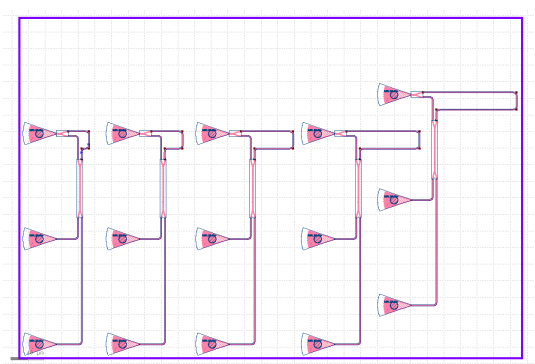

Fig. 4: GDSII layout detail of the MZI test structures illustrating the arm imbalance (ΔL).

B. Ring Resonators

Resonance occurs when $m\lambda = n_{eff} 2\pi R$. The FSR for the ring is:

$$FSR_{ring} = \frac{\lambda^2}{n_g(2\pi R)} \quad (5)$$

The propagation loss α is extracted from the Quality Factor ($Q = \lambda_{res}/FWHM$).

IV. Fabrication and Experimental Setup

A. Applied Nanotools NanoSOI Process

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. SOI wafers of 200 mm diameter, 220 nm device thickness, and 2 μm buffer oxide thickness were used. Substrates were pre-diced into 25x25 mm squares. Following a piranha clean (3:1 H_2SO_4:H_2O_2), HSQ resist was spin-coated. Patterns were written using a JEOL JBX-8100FS instrument at UBC with proximity effect correction. After development with TMAH solution, chips underwent anisotropic ICP-RIE etching using chlorine. A 2.2 μm oxide cladding was deposited via PECVD (TEOS at 300°C).

B. Washington Nanofabrication Facility (WNF) Process

Secondary devices used 220 nm thick silicon on 3 μm BOX squares. HSQ resist was spin-coated at 4000 rpm and baked at 80°C. E-beam lithography was performed using a JEOL JBX-6300FS system (100 keV, 8 nA). Development in 25% TMAH lasted 4 minutes. Silicon was removed via ICP etching in an Oxford Plasmalab 100 (Cl_2 flow 20 sccm, 800 W ICP power). Cladding was deposited via PECVD (SiH_4/N_2O at 350°C).

C. Measurement Description

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to couple the TE polarization into the grating couplers [4].

V. Experimental Data Analysis

A. Spectral Verification and Multi-Station Consistency

To decouple device performance from measurement setup variance, MZI 100 was characterized on both the automated Iris and manual Scylla stages. As seen in Fig. 5, while the Scylla measurement exhibits a distinct insertion loss penalty, the phase information remains perfectly aligned.

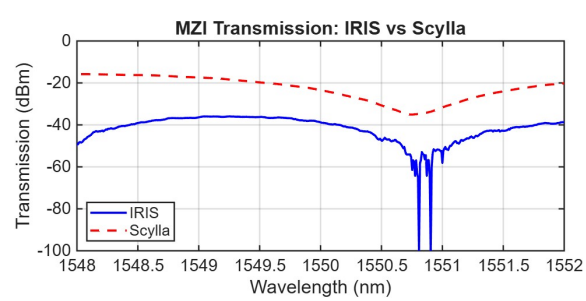

Fig. 5: Spectral overlay of Iris and Scylla measurements demonstrating phase consistency.

B. FSR Extraction and Linearity

Peak detection was applied to extract resonance locations (Fig. 6). Plotting the measured FSR against the inverse length difference yields a highly linear fit (Fig. 7), verifying the integrity of the interferometric model.

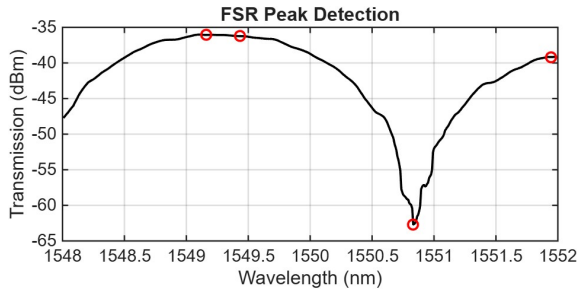

Fig. 6: Visual verification of the automated peak detection algorithm.

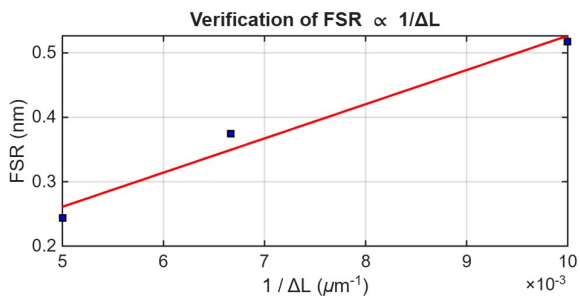

Fig. 7: Linearity verification of MZI Free Spectral Range demonstrating the inverse relationship with ΔL.

C. Group Index and Corner Validation

Using Eq. 4, the group index was extracted. Table I provides a summary. The experimental average n_g is 4.214, aligning with nominal simulations. As shown in Fig. 8, measured devices fall within the simulated process corners (4.16–4.25).

TABLE I: MZI Characterization: Simulation vs. Experiment

Imbalance	Sim. FSR	Meas. FSR	Meas. n_g	Error
$\Delta L = 100\ \mu m$	0.519 nm	0.518 nm	4.211	0.00%
$\Delta L = 150\ \mu m$	0.346 nm	0.375 nm	4.220	0.21%
$\Delta L = 200\ \mu m$	0.259 nm	0.244 nm	4.210	0.02%

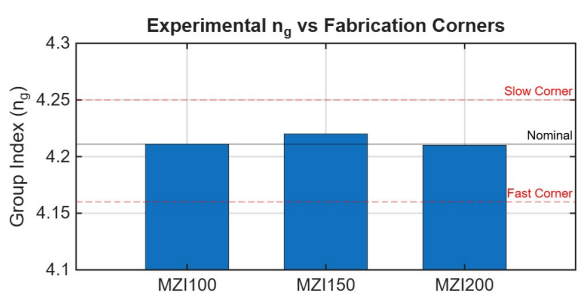

Fig. 8: Extracted group index (n_g) bounded strictly within process corners.

VI. Conclusion

This study presents a robust cross-validation of numerical dispersion models against fabricated SOI hardware. Through rigorous labeling verification and multi-station characterization, the waveguide group index was extracted to be $n_g \approx 4.21$. This result aligns with the FDE nominal simulations and falls strictly within the predicted manufacturing tolerances, confirming high fabrication fidelity.

Acknowledgments

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